

RAJARAM MOHAN BALAJI

+33 7 45 35 66 76 | mohanbalaji1008@gmail.com | linkedin.com/in/mohan1008 | github.com/Mohan790 | 75013 Paris, France

PROFILE

Data Engineering & Cloud Computing master's student who builds production grade systems, not just notebooks. In the past year: shipped a real estate price prediction platform live on Render (Docker, GitHub Actions CI/CD, FastAPI), and built a multi agent AI analyst routing queries across 630+ vector embeddings. Strong in Python, SQL, ETL pipeline design, and containerized deployment looking for a data engineering or ML platform role to keep building systems that run in production, not just in a demo.

KEY TECHNICAL SKILLS

Languages: Python, Java, SQL
Data Engineering: ETL Pipelines, Data Modelling, Data Warehousing, Data Quality
Databases: MySQL, NoSQL Advanced SQL: window functions, CTEs, subqueries
Core CS: OOP, Data Structures & Algorithms, Database Management
Visualisation: Power BI, Plotly, Streamlit KPI dashboards, EDA, data storytelling
Soft Skills: Problem Solving • Team Collaboration • Adaptability • Fast Learner • Analytical Thinking

PROJECTS

- AivancityX : EdTech Admin Dashboard (Ongoing)** | *Python, Streamlit, Plotly*
- Developing a full featured Streamlit admin dashboard for Aivancity as a real college client project
 - Built live KPI widgets and interactive Plotly charts to visualise student and course performance metrics
- OpenHousing : Real Estate Price Prediction Platform** | *Python, FastAPI, scikit-learn, Docker, GitHub Actions* | github.com/Mohan790/openhousing
- Built an end to end ML pipeline automated ETL, a Random Forest model ($R^2 = 0.88$, $MAE \approx \$2,000$) deployed live on Render with Docker and a GitHub Actions CI/CD pipeline
 - Developed a FastAPI REST API with a real time web interface, and shipped per IP rate limiting after a community reported production security review. **Live link :** <https://openhousing-8q7c.onrender.com/app/>
- Enterprise AI Data Analyst** | *Python, LangGraph, Qdrant, FastAPI, Docker* | github.com/Mohan790/enterprise-ai-analyst
- Built a multi modal ReAct agent (LangGraph + Llama 3.1) routing business questions between SQL and semantic vector search over 10 financial reports
 - Engineered a Vector ETL pipeline: semantically chunked documents, embedded with HuggingFace, stored 632+ vectors in a Dockerized Qdrant database
- FreshCheck AI** | *Python, PyTorch, Streamlit, MySQL* | github.com/Mohan790/food-quality-inspector
- Fine tuned MobileNetV2 on 13,000+ real world food images, achieving 99.01% validation accuracy with a custom PyTorch training loop and Grad CAM explainability
 - Deployed a live Streamlit app with real-time Fresh/Rotten classification, confidence charts, and heatmap overlays

PROFESSIONAL EXPERIENCE

- Web Development Intern : Scorpleo Technologies** Sep 2024 – Dec 2024 | Coimbatore, India
- Shipped front-end and back-end features using HTML5, CSS, Python, and SQL directly on live client projects
 - Integrated dynamic SQL-driven data retrieval into production web interfaces; contributed to code reviews and Git-based collaborative workflows
- Trainee Associate Healthcare IT : Baruch Business Solutions** Aug 2023 – Jul 2024 | Coimbatore, India
- Managed daily processing of patient records across healthcare information systems, maintaining data accuracy and regulatory compliance
 - Wrote and optimised SQL queries to extract, clean, and validate patient datasets, reducing manual reporting effort for the team

EDUCATION

MSc Data Engineering & Cloud Computing — Aivancity, Paris	Present
BSc Information Technology — Dr. N.G.P. Arts & Science College	2020 – 2023

CERTIFICATIONS

- Master Data Management for Beginners (Tcs ion) • Big Data Fundamentals (Cognitive Class, IBM) • Artificial Intelligence (Infosys Springboard)
- MySQL & PHP Development (Spoken Tutorial)

LANGUAGES

Tamil: Native • English: Professional (C1) • French: Beginner (B1)